LAPORAN TUGAS PERTEMUAN 4

PROGRAMMING LANGUAGE PRAGMATICS

FINITE AUTOMATA: REGULAR EXPRESSION KE NFA DAN NFA KE DFA

Laporan Ini Disusun Untuk Memenuhi Tugas Kuliah

Mata Kuliah : Programming Language Pragmatics



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BAB I PENDAHULUAN

1.1 Soal

Lima regular expression dari Latihan 1 pertemuan 3 dicari contoh string-nya, lalu diubah menjadi NFA, kemudian NFA tersebut diubah menjadi DFA:

- P → `x(y)+z`, alfabet {x, y, z}
- Q → `m | nr*`, alfabet {m, n, r}
- R → `(p | q)*r`, alfabet {p, q, r}
- T → `t*(u | v)`, alfabet {t, u, v}
- W → `x(y | z)+`, alfabet {x, y, z}

1.2 Latihan 1: String yang Dihasilkan

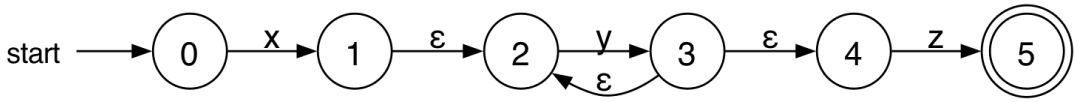
- P = {xyz, xyyz, xyyyz, ...}
- Q = {m, n, nr, nrr, ...}
- R = {r, pr, qr, pqr, ...}
- T = {u, v, tu, tv, ...}
- W = {xy, xz, xyz, xzy, ...}

1.3 Cara Pengerjaan

- Regex ke NFA memakai aturan di slide *RegEx to NFA*: NFA untuk simbol, concatenation (state akhir N(s) digabung dengan state awal N(t)), union, Kleene closure, dan positive closure. Penomoran state mengikuti contoh (a|b)*abb di slide, dari kiri ke kanan dan cabang atas lebih dulu.
- NFA ke DFA memakai *ε-closure* dan *move* seperti di slide *Conversion of an NFA to a DFA*. Kalau hasil *move* sudah pernah dihitung, *ε-closure*-nya tidak ditulis ulang.
- ∅ berarti tidak ada transisi. Pada DFA, ∅ adalah dead state yang tidak digambar di diagram.
- Semua NFA dan DFA dicek dengan program: untuk setiap string sampai panjang 6 atas alfabetnya, NFA dan DFA menerima string yang sama persis dengan regex aslinya.

BAB II P → x(y)+z

2.1 Regex ke NFA



Gambar 2.1: NFA untuk P → x(y)+z

Tabel 2.1: Tabel transisi NFA P

State	x	y	z	ε
0	{1}	∅	∅	∅
1	∅	∅	∅	{2}
2	∅	{3}	∅	∅
3	∅	∅	∅	{2, 4}
4	∅	∅	{5}	∅
5	∅	∅	∅	∅

State awal 0, state final 5.

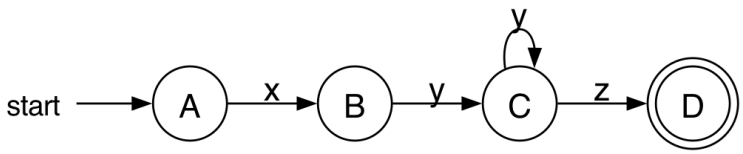
2.2 NFA ke DFA

- *ε-closure* (0) = {0} → **A**
- *move* [A, x] = {1}
- *ε-closure* ([A, x]) = *ε-closure* ({1}) = {1, 2} → **B**
- *move* [A, y] = *move* [A, z] = ∅
- *move* [B, y] = {3}
- *ε-closure* ([B, y]) = *ε-closure* ({3}) = {2, 3, 4} → **C**
- *move* [B, x] = *move* [B, z] = ∅
- *move* [C, y] = {3} → **C**
- *move* [C, z] = {5}
- *ε-closure* ([C, z]) = *ε-closure* ({5}) = {5} → **D**
- *move* [C, x] = ∅
- *move* [D, x] = *move* [D, y] = *move* [D, z] = ∅

Tabel 2.2: Tabel transisi DFA P

State NFA	State DFA	x	y	z
{0}	A	B	∅	∅
{1, 2}	B	∅	C	∅
{2, 3, 4}	C	∅	C	D
{5}	**D**	∅	∅	∅

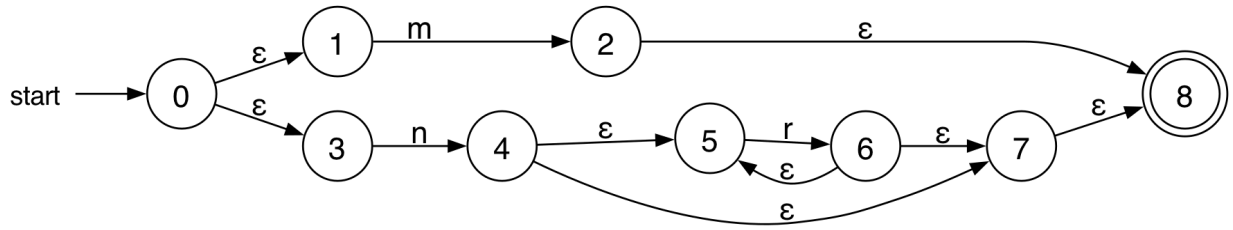
State awal A, state final D (dicetak tebal) karena memuat state final NFA 5.



Gambar 2.2: DFA untuk P → x(y)+z

BAB III Q → `m | nr*`

3.1 Regex ke NFA



*Gambar 3.1: NFA untuk Q → m | nr**

Tabel 3.1: Tabel transisi NFA Q

State	m	n	r	ε
0	∅	∅	∅	{1, 3}
1	{2}	∅	∅	∅
2	∅	∅	∅	{8}
3	∅	{4}	∅	∅
4	∅	∅	∅	{5, 7}
5	∅	∅	{6}	∅
6	∅	∅	∅	{5, 7}
7	∅	∅	∅	{8}
8	∅	∅	∅	∅

State awal 0, state final 8.

3.2 NFA ke DFA

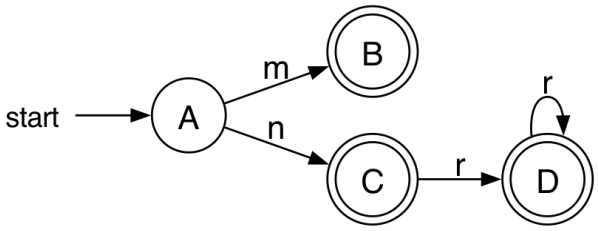
- *ε-closure* (0) = {0, 1, 3} → **A**
- *move* [A, m] = {2}
- *ε-closure* ([A, m]) = *ε-closure* ({2}) = {2, 8} → **B**
- *move* [A, n] = {4}
- *ε-closure* ([A, n]) = *ε-closure* ({4}) = {4, 5, 7, 8} → **C**
- *move* [A, r] = ∅
- *move* [B, m] = *move* [B, n] = *move* [B, r] = ∅
- *move* [C, r] = {6}
- *ε-closure* ([C, r]) = *ε-closure* ({6}) = {5, 6, 7, 8} → **D**
- *move* [C, m] = *move* [C, n] = ∅

- *move* [D, r] = {6} → **D**
- *move* [D, m] = *move* [D, n] = ∅

Tabel 3.2: Tabel transisi DFA Q

State NFA	State DFA	m	n	r
{0, 1, 3}	A	B	C	∅
{2, 8}	**B**	∅	∅	∅
{4, 5, 7, 8}	**C**	∅	∅	D
{5, 6, 7, 8}	**D**	∅	∅	D

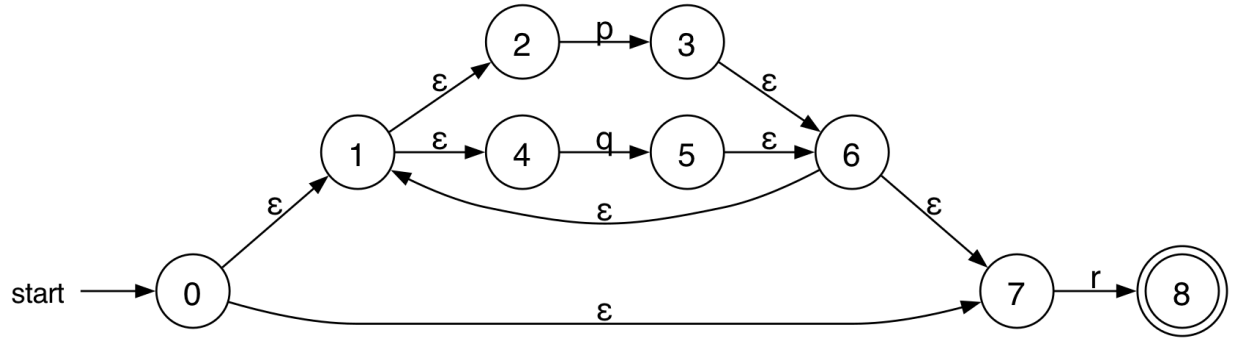
State awal A, state final B, C, D (dicetak tebal) karena memuat state final NFA 8.



*Gambar 3.2: DFA untuk Q → m | nr**

BAB IV R → (p | q)*r

4.1 Regex ke NFA



*Gambar 4.1: NFA untuk R → (p | q)*r*

Tabel 4.1: Tabel transisi NFA R

State	p	q	r	ε
0	∅	∅	∅	{1, 7}
1	∅	∅	∅	{2, 4}
2	{3}	∅	∅	∅
3	∅	∅	∅	{6}
4	∅	{5}	∅	∅
5	∅	∅	∅	{6}
6	∅	∅	∅	{1, 7}
7	∅	∅	{8}	∅
8	∅	∅	∅	∅

State awal 0, state final 8.

4.2 NFA ke DFA

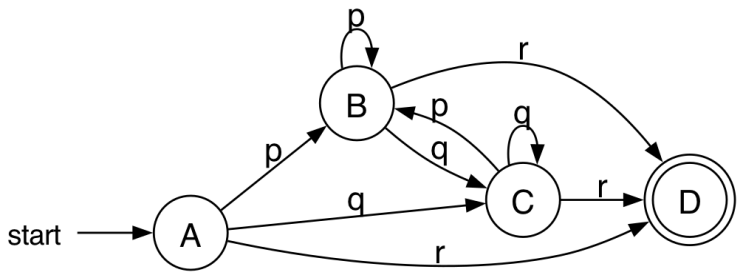
- *ε-closure* (0) = {0, 1, 2, 4, 7} → **A**
- *move* [A, p] = {3}
- *ε-closure* ([A, p]) = *ε-closure* ({3}) = {1, 2, 3, 4, 6, 7} → **B**
- *move* [A, q] = {5}
- *ε-closure* ([A, q]) = *ε-closure* ({5}) = {1, 2, 4, 5, 6, 7} → **C**
- *move* [A, r] = {8}
- *ε-closure* ([A, r]) = *ε-closure* ({8}) = {8} → **D**
- *move* [B, p] = {3} → **B**

- *move* [B, q] = {5} → **C**
- *move* [B, r] = {8} → **D**
- *move* [C, p] = {3} → **B**
- *move* [C, q] = {5} → **C**
- *move* [C, r] = {8} → **D**
- *move* [D, p] = *move* [D, q] = *move* [D, r] = ∅

Tabel 4.2: Tabel transisi DFA R

State NFA	**State DFA**	**p**	**q**	**r**
{0, 1, 2, 4, 7}	A	B	C	D
{1, 2, 3, 4, 6, 7}	B	B	C	D
{1, 2, 4, 5, 6, 7}	C	B	C	D
{8}	**D**	∅	∅	∅

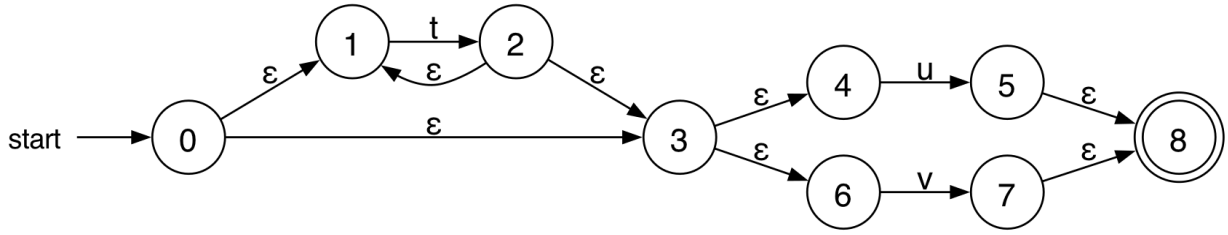
State awal A, state final D (dicetak tebal) karena memuat state final NFA 8.



*Gambar 4.2: DFA untuk R → (p | q)*r*

BAB V T → t*(u | v)

5.1 Regex ke NFA



Gambar 5.1: NFA untuk T → t(u | v)*

Tabel 5.1: Tabel transisi NFA T

State	t	u	v	ε
0	∅	∅	∅	{1, 3}
1	{2}	∅	∅	∅
2	∅	∅	∅	{1, 3}
3	∅	∅	∅	{4, 6}
4	∅	{5}	∅	∅
5	∅	∅	∅	{8}
6	∅	∅	{7}	∅
7	∅	∅	∅	{8}
8	∅	∅	∅	∅

State awal 0, state final 8.

5.2 NFA ke DFA

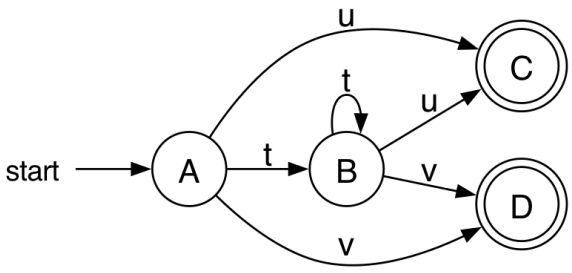
- *ε-closure* (0) = {0, 1, 3, 4, 6} → **A**
- *move* [A, t] = {2}
- *ε-closure* ([A, t]) = *ε-closure* ({2}) = {1, 2, 3, 4, 6} → **B**
- *move* [A, u] = {5}
- *ε-closure* ([A, u]) = *ε-closure* ({5}) = {5, 8} → **C**
- *move* [A, v] = {7}
- *ε-closure* ([A, v]) = *ε-closure* ({7}) = {7, 8} → **D**
- *move* [B, t] = {2} → **B**
- *move* [B, u] = {5} → **C**
- *move* [B, v] = {7} → **D**

- *move* [C, t] = *move* [C, u] = *move* [C, v] = ∅
- *move* [D, t] = *move* [D, u] = *move* [D, v] = ∅

Tabel 5.2: Tabel transisi DFA T

State NFA	State DFA	t	u	v
{0, 1, 3, 4, 6}	A	B	C	D
{1, 2, 3, 4, 6}	B	B	C	D
{5, 8}	**C**	∅	∅	∅
{7, 8}	**D**	∅	∅	∅

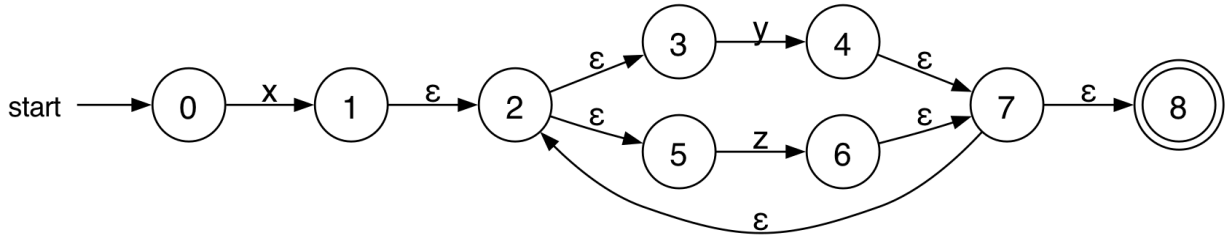
State awal A, state final C, D (dicetak tebal) karena memuat state final NFA 8.



Gambar 5.2: DFA untuk T → t(u | v)*

BAB VI W → x(y | z)+

6.1 Regex ke NFA



Gambar 6.1: NFA untuk W → x(y | z)+

Tabel 6.1: Tabel transisi NFA W

State	x	y	z	ε
0	{1}	∅	∅	∅
1	∅	∅	∅	{2}
2	∅	∅	∅	{3, 5}
3	∅	{4}	∅	∅
4	∅	∅	∅	{7}
5	∅	∅	{6}	∅
6	∅	∅	∅	{7}
7	∅	∅	∅	{2, 8}
8	∅	∅	∅	∅

State awal 0, state final 8.

6.2 NFA ke DFA

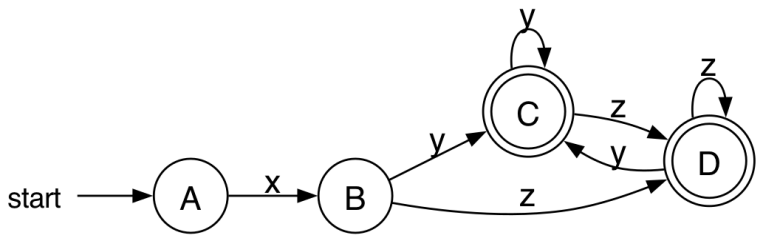
- *ε-closure* (0) = {0} → **A**
- *move* [A, x] = {1}
- *ε-closure* ([A, x]) = *ε-closure* ({1}) = {1, 2, 3, 5} → **B**
- *move* [A, y] = *move* [A, z] = ∅
- *move* [B, y] = {4}
- *ε-closure* ([B, y]) = *ε-closure* ({4}) = {2, 3, 4, 5, 7, 8} → **C**
- *move* [B, z] = {6}
- *ε-closure* ([B, z]) = *ε-closure* ({6}) = {2, 3, 5, 6, 7, 8} → **D**
- *move* [B, x] = ∅
- *move* [C, y] = {4} → **C**

- *move* [C, z] = {6} → **D**
- *move* [C, x] = ∅
- *move* [D, y] = {4} → **C**
- *move* [D, z] = {6} → **D**
- *move* [D, x] = ∅

Tabel 6.2: Tabel transisi DFA W

State NFA	**State DFA**	**x**	**y**	**z**
{0}	A	B	∅	∅
{1, 2, 3, 5}	B	∅	C	D
{2, 3, 4, 5, 7, 8}	**C**	∅	C	D
{2, 3, 5, 6, 7, 8}	**D**	∅	C	D

State awal A, state final C, D (dicetak tebal) karena memuat state final NFA 8.



Gambar 6.2: DFA untuk W → x(y | z)+

BAB VII RINGKASAN

Tabel 7.1: Ringkasan hasil konversi

Nama	Regex	State NFA	State DFA	State final DFA
P	`x(y)+z`	6	4	D
Q	`m | nr*`	9	4	B, C, D
R	`(p | q)*r`	9	4	D
T	`t*(u | v)`	9	4	C, D
W	`x(y | z)+`	9	4	C, D